FRTN30 Assignment 1

August Hermansson au3113he-s
Gustaf Utterström gu4303ut-s

April 24, 2026

1

1.1 Degree centrality

The out-degree and in-degree centralities were computed with

$$C_{\text{out}}(i) = \sum_j A_{ij}, \quad C_{\text{in}}(i) = \sum_j A_{ji}.$$

where A is the respective country's weight matrix. These were the results for swe2000 and idn2000,
Sweden (2000):

- Top 3 out-degree sectors:
  - Other Business Activities (257819.52)
  - Real estate activities (115105.81)
  - Wholesale & retail trade; repairs (105814.99)
- Top 3 in-degree sectors:
  - Radio (142431.72)
  - Motor vehicles (141428.93)
  - Other Business Activities (135142.95)

Indonesia (2000):

- Top 3 out-degree sectors:
  - Wholesale & retail trade; repairs (1.78×10^8)
  - Agriculture (1.75×10^8)
  - Mining and quarrying (energy) (1.10×10^8)
- Top 3 in-degree sectors:
  - Food products (2.14×10^8)
  - Construction (1.51×10^8)
  - Wholesale & retail trade; repairs (1.13×10^8)

Here we see that Sweden has higher centrality in service sectors and Indonesia in production sectors.

1.2 Eigenvector centrality

Eigenvector centrality is defined by

$$A^T x = \lambda x,$$

where x is the centrality vector, and A is the same as in previous exercise. The three largest eigenvalues with corresponding eignevectors are what's asked for. This eigenvector problem was solved with numpy as well as networkx to compute largest connected component, which is important since eigenvector centrality is not unique unless the graph is strongly connected (Perron-Frobenius). This gave us the following results, **Sweden (2000):**

- Motor vehicles (0.5183)
- Radio (0.4288)
- Other Business Activities (0.3402)

Indonesia (2000):

- Food products (0.9108)
- Hotels & restaurants (0.2767)
- Agriculture (0.2735)

We find similar sectors having high eigenvector centrality as well as degree centrality.

1.3 Katz Centrality

Katz centrality is given by

$$x = \beta(I - \alpha A^T)^{-1}\mu,$$

with $\alpha = \frac{1-\beta}{\lambda_{\max}}$ and given $\beta = 0.15$ and the two cases:

1. $\mu = \mathbf{1}$ (uniform importance),
2. μ concentrated on the "Wholesale & retail trade" sector.

as the centrality could be expressed on a closed form we computed it straight away (numpy) which gave the following,

Case 1: Uniform μ

- Sweden:
  - Motor vehicles (2.2317)
  - Radio (2.0082)
  - Other Business Activities (1.6989)
- Indonesia:
  - Food products (3.0172)
  - Hotels & restaurants (1.0233)

- Agriculture (1.0128)

Case 2: Targeted μ (Wholesale & retail trade)

- Sweden:
  - Wholesale & retail trade; repairs (0.2071)
  - Motor vehicles (0.1087)
  - Radio (0.0903)
- Indonesia:
  - Food products (0.4396)
  - Wholesale & retail trade; repairs (0.1883)
  - Hotels & restaurants (0.1479)

We see that with a uniform μ we get the same results as for the eigenvector centrality whereas when focused on one sector this sector also had a higher centrality.

2

2.1 a)

To compute the PageRank we started with the fomrula:

$$\mathbf{z} = (1-\beta)\mathbf{P}^T\mathbf{z} + \beta\mu \tag{1}$$

With $\mathbf{z} \in R^{nx1}$ as the centrality measure. β represents the probability of a random jump, or not following a link. Hence, $1-\beta$ the probability of following a link. Furthermore, $\mathbf{P}^T \in R^{nxn}$ is the transition matrix that describes the probability of flow from node j to node i. $\beta\mu$ represent the probability of a complete random jump, same probability for all nodes due to normalization.

To find the centrality vector $\mathbf{z}$ we use an iterative approach since the direct formula is computationally expensive. By using the iteration approach we repeatedly propagate importance through the network, and add some randomness with the last term in equation 1. In that way we iterate forward to the converging solution.

To construct the matrix $\mathbf{P^T}$ we normalize each row of the given matrix $\mathbf{w}$ by computing $\mathbf{Dw}$, $\mathbf{D} = diag(\frac{1}{rowsum_i})$. By this normalization, we can now interpret the matrix $\mathbf{P^T}$ as a transition matrix. Notable was that in the given dataset, one rowsum was equal to zero. To handle this, we just set this value to 1 in the vector $row-sum$, this does not change the function of the graph, but just avoids division by zero in the matrix $\mathbf{D}$. Before starting the iteration, we initially used $\mathbf{z} = \mathbb{1}$, which means that we assumed that all nodes are equally important and will converge to the right solution. The top-ranked nodes according to the PageRank vector are given by

Node	PageRank-value
2	0.0178
1	0.0156
112	0.0119
9	0.0103
26	0.0091

Table 1: Top 5 PageRank nodes

In Table 1 the PageRank-values for all 5 nodes are similar. Node 2, is the greatest, and therefore has many followers and, more specifically, many "important" followers. The other nodes have smaller values, and influence, but similar values and are still the base of the network since they have considerable centrality.

2.2 b)

In this exercise, the discrete-time consensus algorithm $\mathbf{z} = \mathbf{P}\mathbf{z}$, with $\mathbf{P}$ constructed the same way as described above, was calculated. The algorithm computes how an opinion is spread through a network, and each nodes opinion is updated by computing a weighted average of its neighbors (accounts its follows) opinions. In the exercise, we initialized all nodes to have opinion, $\mathbf{z} = 0.5$ expect two stubborn nodes, with opinion 1 respectively 0. We then iterated the formula above to see how the opinions are spread throughout the network while observing the changing opinion for 5 nodes.

In Figure 1 we can see that the stubborn nodes keep their opinion all the time, while the 3 others converge to an opinion between 0 and 1. This behavior may be explained by the fact that the opinions are computed by a weighted average, from neighboring nodes. We can state that node 112 is more influenced by node 9 and node 5001 and 6001 by node 26. In the previous exercise the PageRank algorithm showed that node 9 is more central than 26. With this in mind, the intuition may have been that more nodes should have converged to the opinion of node 9. However, the consensus algorithm is not mainly based on centrality, but on how strongly and how often other nodes weight "you" in their updates. For instance, node 26 may have less followers than 9, but each follower may follow fewer accounts, and therefore the weight of node 26 opinion may dominate.

In Figure 2 we can observe a similar behavior as described above. The picked stubborn nodes have even higher centrality in this plot, and therefore the other nodes opinions converge to either 0 or 1. The importance of centrality for the spread of the opinion is also shown in Figure 3. In this plot, we can see that all plotted nodes converge to the same opinion as the more central node, node 1, in the network rather than a completely random stubborn node, node 4568.

Finally, we can look at Figure 4, which compares the centrality of node 1 and node 2. In this plot, we can observe that the most central node, node 2, does not dominate the spread of opinion, in contrast, Figure 2. In Figure 4 we have chosen other nodes to observe how their opinion changed over time than in Figure 2. This observation claims that the opinion spread is based both on the centrality of the stubborn nodes but also on how strongly and often other nodes weight "you" in their updates.

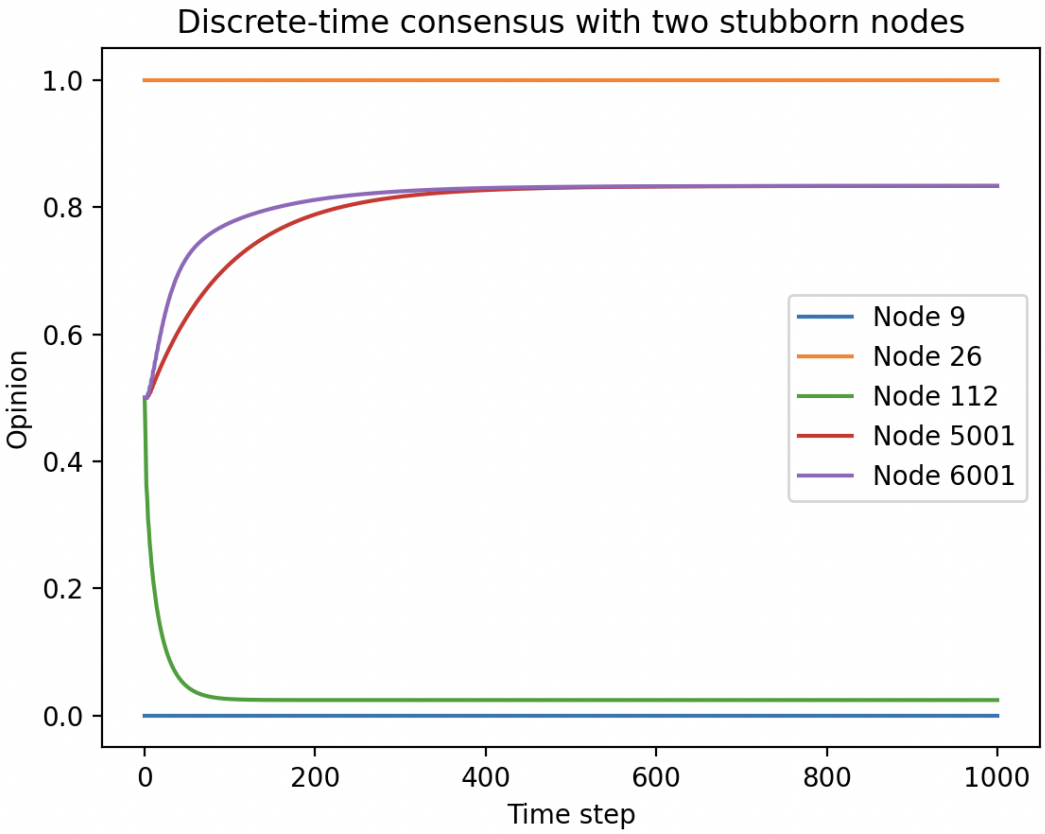


Figure 1: With stubborn nodes: Node 9=1 and Node 26=0

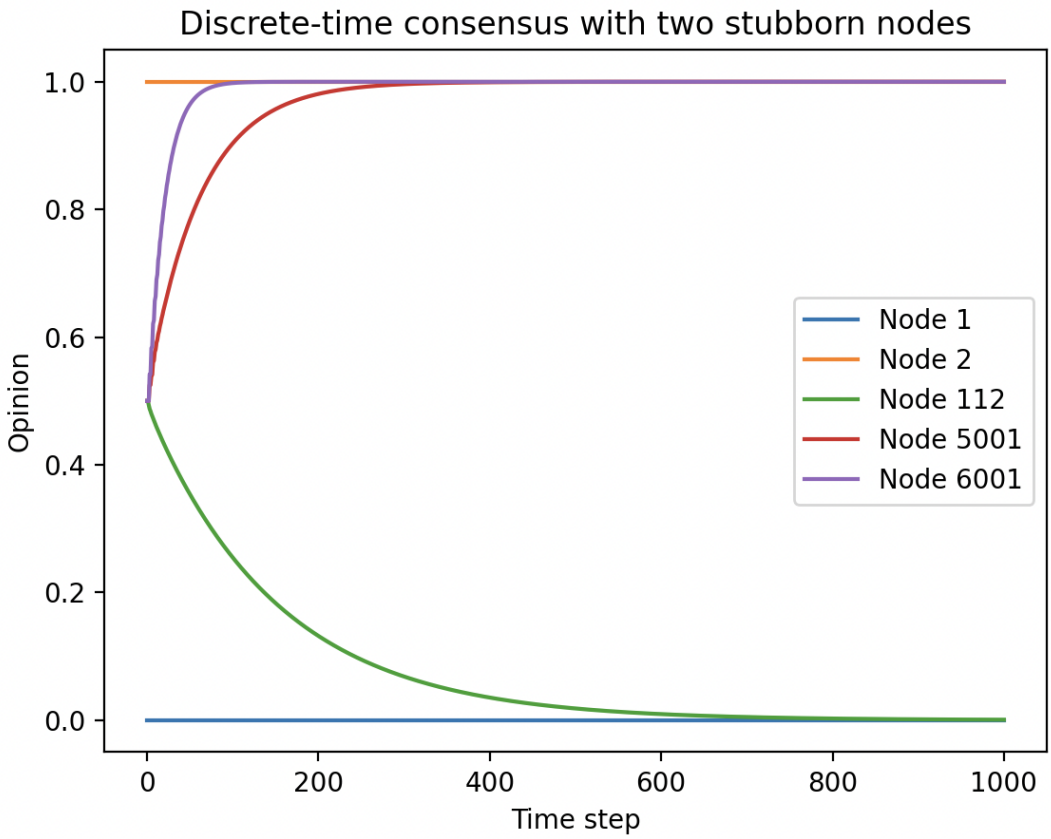


Figure 2: With stubborn nodes: Node 1=0 and Node 2=1

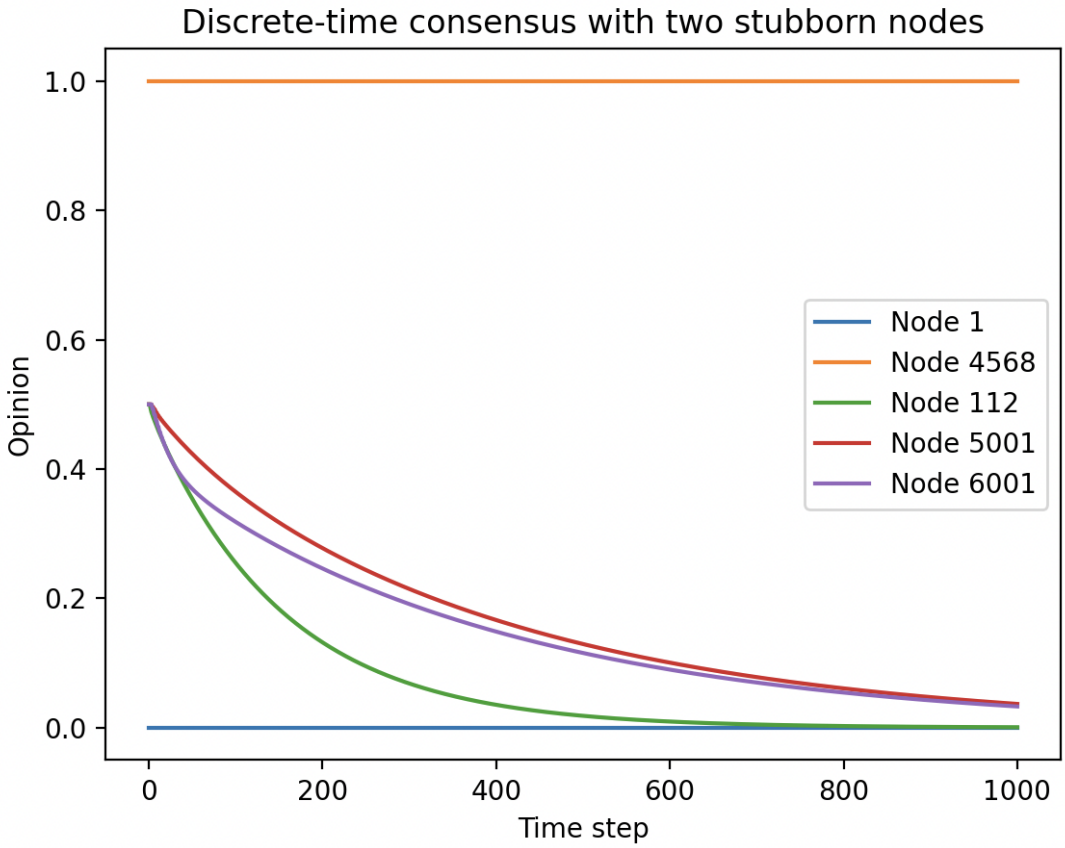


Figure 3: With stubborn nodes: Node 1=0 and Node 4568=1

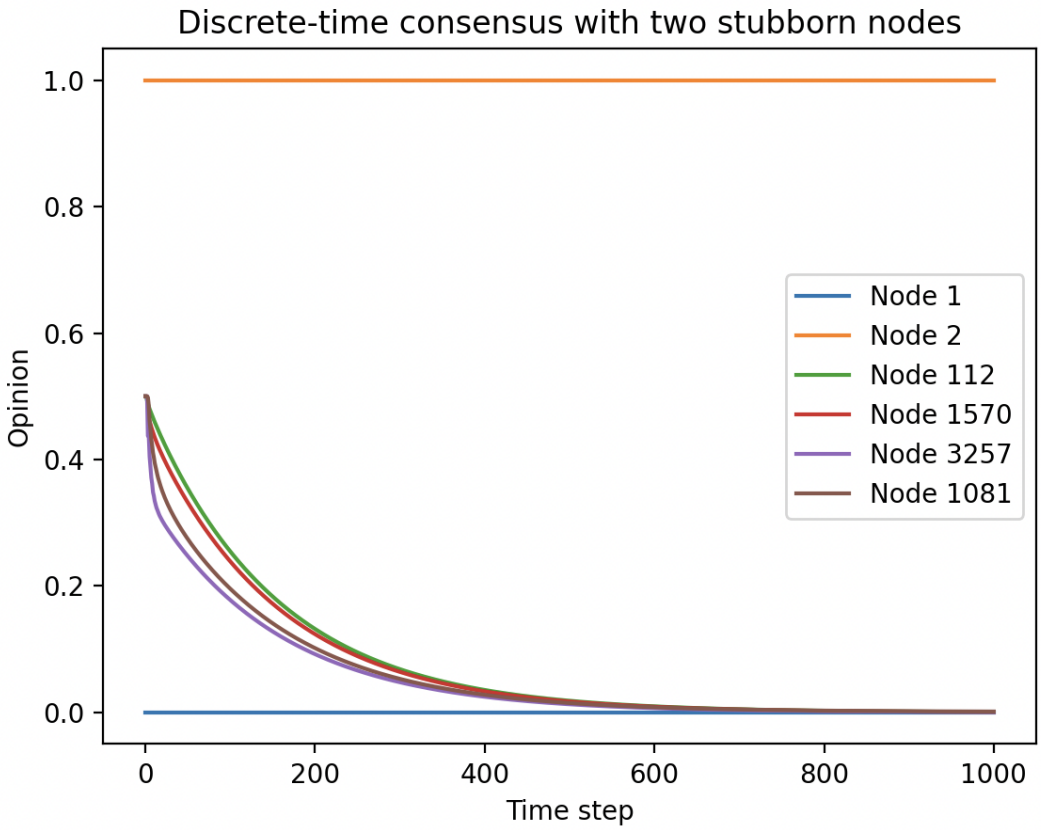


Figure 4: With stubborn nodes: Node 1=0 and Node 2=1